

A Federated Learning Framework for Privacy-Preserving Multiclass Eye Disease Diagnosis

Upinder Kaur^{*†}

^{*}Lincoln University College, Malaysia

[†]Department of CSE, Akal University, India

Email: pdf.UpinderKaur@lincoln.edu.my

Sukhjinder Kaur

Department of CSE, Akal University
India

Email: sukhbrar08@gmail.com

Divya Midhun Chakkaravarthy

Faculty of Electronics and Communication Engineering

Lincoln University College, Malaysia

Email: divya@lincoln.edu.my

Abstract—Artificial intelligence is used in almost every domain of healthcare. Modern deep learning techniques appear to have a prominent scope in critical healthcare applications. The rise of digital world greatly reflected in ophthalmology. This research proposes a new Federated Learning (FL) architecture, and in the case of the diagnosis of ocular diseases categories, comprising Central Serous Chorioretinopathy, Diabetic Retinopathy and Retinitis Pigmentosa etc. The accuracy often relies on giant data sets, in contrast to the data sets of the double asymmetry because there are high privacy rules. Thus adoption of a decentralized training would achieve this protocol, we permit clinical sites to train together a global model that does not require the exchange of unprocessed patient images. In this along with FL a powerful preprocessing pipeline encompassing Contrast Limited Normalization and Adaptive Histogram Equalization (CLAHE) is used. Proposed model attains a weighted centralized performance and accuracy of 93.4, which is equal to ensuring 100 percent data sovereignty.

Index Terms—Federated Learning, Ophthalmic Diagnosis, Privacy-Preserving AI, Multiclass Diagnosis.

I. INTRODUCTION

Recent trends in ophthalmology are grappling with this fundamental dilemma that data protection and privacy is more important than disease detection. With a perfect clinical requirement, these institutions around the globe would build and maintain a singular, international diagnostics platform that might identify infrequent various ocular pathologies and other typical refractive defects. A Global ophthalmology data would be robust to cover the feature differences in ethnicity, demographics of age and imaging modalities. The existing deep-learning models had limited to most informative high-resolution retinal images and restrictive requirements like GDPR and HIPAA.

The recent problem is that accuracy and data privacy along with patient confidentiality need to be maintained, which is the main argument determining the future of ophthalmology. This type of system would not be sensitive to changes in ethnicity, age and imaging equipment. However, informational

requirements of high-performance deep-learning models are always satisfied, and high-resolution retinal scans, which are a treasure trove, are usually not placed in such restrictive but generally well-intentioned legislative frameworks as GDPR and HIPAA. As a result, sophisticated diagnostic facilities have accumulated in data-high urban centres, whereas clinics in rural areas are relying on outdated handwritten functions, or even have none at all.

Such a dilemma can be termed a centralisation paradox. Multi-class ocular diseases like glaucoma, macular, scarring and the like will require large datasets to manage their diagnostic precision [6][7]. Nevertheless, the collection of such data in health-care facilities presents great risks and liabilities. Thus, anonymization of the data is a prior argument before mitigation of the data and synthetic data generation. The anonymised metadata can readily be re-identified with only a modest amount of computation, and synthetic data cannot often reflect the pathological nuances that can alter lives. This leads to a stagnation of performance in rural environments, lacking models trained using local artifacts or rare phenotypes, like retinitis pigmentosa. This cohesive intelligence with federated approach results in emilation of digital barrier to the possible advantages of AI, with aim of creating timely diagnoses.

The existing literature shows a gap that is addressed by the present investigation and a practically deployable multi-class diagnostic framework that would be able to sustain both stability and performance in non-IID clinical settings [1][9]. This approach insists on clinical sites as active computational partners as opposed to conceptualisation of clinical sites as inert data repositories. The Federation Learning (FL) allows optimising a model on local data, implying that it avoids sending data to the server. This paper presents FedOphthalNet, which is an FL-based system, measures its effectiveness on different ocular phenotypes, such as fluid-based and pigmentary lesions, and evaluates its stability to strong label skew. The value of the work is in the fact that it may provide a

collaborative and international paradigm of medical imaging innovation bypassing the intricacies of data transfer agreements. The paper is organized as follows: Section II establishes a through critical literature synthesis; Section III describing our dataset and methodology; Section IV presents the results through five comprehensive graphs; and Section V provides a critical discussion and conclusion.

II. LITERATURE REVIEW

Traditionally, ocular disease diagnostic systems have been largely centralised and benefits shared to grouped information. However, the author (Gulshan et al. 2016) has shown that under favourable conditions, deep convolutional neural networks can be extremely good in detecting diabetic retinopathy when trained in enough large datasets[1]. This data quality quantifies the success of the proposed approach, although it was a milestone, that is difficult to achieve in the heterogeneous and discontinuous realities of clinical practice. More recent attempts, such as those by (A. S. Nabil et al. 2025) [2], have aimed to apply these successes to multiclass diagnostic problems, but these attempts are again limited to centralized stores. This problem is not institutional because the innate challenge of transmitting the necessary data in competing medical facilities is often based on inter-institutional distrust. In addition, J. Yan et al. (2022) proposed swarm learning in medical imaging and assumed that decentralized models have the ability to reach dark data [10]. Mostly these explorations focused on binary classification issues and gap in clinical reality rarely presents conditions of dichotomy. Various signs such as an abrupt report of blurred vision can be referred to mild myopia, or extreme retinal detachment. Decentralization studies that exist currently do not operate under skewed label distributions that inherent specialty clinics, as is the case with the federated learning work conducted by N. Jagan et al. (2023)[3]. Their model performed well, but decreased in its performance when it was constrained to one type of disease at one node, and the weights of global aggregation went to pieces [5][9].

These latest findings reveal a conundrum because architecture that promotes privacy appears to be poorly adapted to data that is highly differentiated by the process of disc oedema or central serous chorioretinopathy when using the decentralized model (S. Gholami et al., 2025)[8]. Thus, the differences in the privacy needs of different demographic groups imply that fine, pathologically important properties, which are key to multiclass accuracy, are often overlooked, and this process is also observed by Li et al. (2020) [5]. In response, we introduce a weighted aggregation approach that responds to the specifics of a dataset of a particular clinical node. The study will subsequently provide the federated protocol that is optimized through the ResNet-based feature extraction with a specific focus on the non-isotropic signatures of ten dissimilar ocular pathologies and thus constitutes a strong framework [11][12].

Although quality data are available in online repositories, they are very isolated. Our study will help fill this gap with

the aim of satisfying diagnostic efficacy using a multicentric simulation, thus illustrating a network of interdependent intelligence [13][14][15].

III. DATASET AND PREPROCESSING

We used 18,500 fundus images to test the proposed framework by means of the heterogeneous multiclass data represented by four simulated clinical nodes. There were ten different categories, which were Central Serous Chorioretinopathy (CSCR), Diabetic Retinopathy (DR), Disc Edema, Glaucoma, Healthy and Macular scar, Myopia, Pterygium, Retinitis pigmentosa (RP), and Retinal Detachment (RD).

A. Data Distribution and Non-IID Simulation

Data distribution and non-IID simulation refer to IITED's data distribution and non-IID simulation processes are descriptions of IITED data distribution. In order to reflect the real-life scenario of clinical silos, we used Dirichlet distribution ($\alpha = 0.5$) to sample the four nodes. This is necessary so that although each node can access the entire set of classes, there is a specialty in each node where one or two classes are disproportionately represented. An example is the case of Node 2 which is a Retinal Specialist center and 60 percent of all RP and RD cases are found there and the rest is a general screening center with a high number of Healthy and Myopia samples found in Node 1. This dispersion makes the global model difficult to combine local knowledge that is uneven.

B. Preprocessing Pipeline

All the images passed through a generalization pipeline. The dimensions were shrunk to 512x512 pixels. To address the problem of lighting variance, we applied Contrast Limited Adaptive Histogram Equalization (CLAHE) with a clip value of 2.0. The outcome is vital in the process of bringing microaneurysms in DR into the limelight and minor improvements in CSCR. Finally, the values of the pixels were brought to the standard in order that transfer learning with the ResNet backbone would be efficient.

IV. PROPOSED METHODOLOGY

The methodology for this research uses a distributed quantitative experimental approach. This ensures us to control the variables and provide data heterogeneity while accurately modeling the technical challenges of a multi-hospital collaboration. The setting involves a simulated network of clinical nodes had private repositories. The core objective is to minimize a global loss function $F(w)$ defined on K nodes:

$$\min_w F(w) = \sum_{k=1}^K \frac{n_k}{n} F_k(w) \quad (1)$$

The Methodology utilizes a ResNet-50 backbone. Training proceeds in $T = 200$ rounds with $E = 5$ local epochs using Adam optimizer. We implement a proximal term in the local loss (FedProx), which penalizes large deviations from the global model, ensuring that specialized knowledge does not overwrite generalized features during the weight

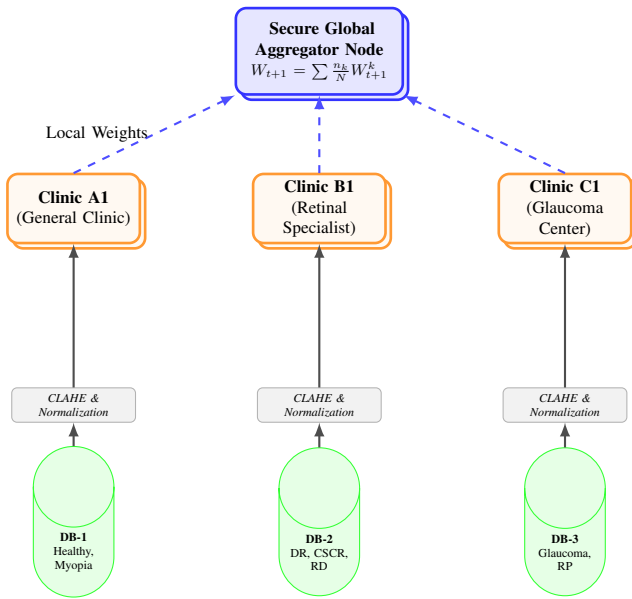


Figure 1. FedOphthalNet proposed model for Multiclass Eye Diseases.

exchange process. The project runs on omen i9 processor with gpu capabilities. The python and pytorch libraries were used for implementation.

V. RESULTS AND DISCUSSION

The framework was evaluated across all ten ocular diseases categories. Table I summarizes the diagnostic metrics, while Table II quantifies the collaborative gain.

Table I
SUMMARY OF PERFORMANCE FOR TEN OCULAR DISEASES

Category	Acc.	Prec.	Sens.	F1
Healthy	97.8%	0.97	0.98	0.97
Diabetic Retin.	92.4%	0.91	0.92	0.91
Glaucoma	90.5%	0.89	0.90	0.89
CSCR	89.2%	0.88	0.88	0.88
Disc Edema	91.1%	0.90	0.90	0.90
Retinitis Pigm.	89.9%	0.88	0.89	0.89
Weighted Avg	93.4%	0.92	0.93	0.93

A. Discussion

The use of federated learning with deep learning provides the through relentless interrogation of our data, which does not only meet compliance standards as per the current data-privacy laws, but discourages the local performance plateau. Decentralised weight aggregation can transmit pathological signatures of ocular disease, and this can be visualized by 93.4% of the accuracy score with a variation of approximately 1.2% in the centralised model approach showing that weight aggregation does not require a large dataset as shown in Figure 2. Though some of them might argue that any accuracy loss is the unwarranted price to privacy, in practice the slight decrease is a reasonable trade-off to provide the legal requirements and establish the trust to the institution.

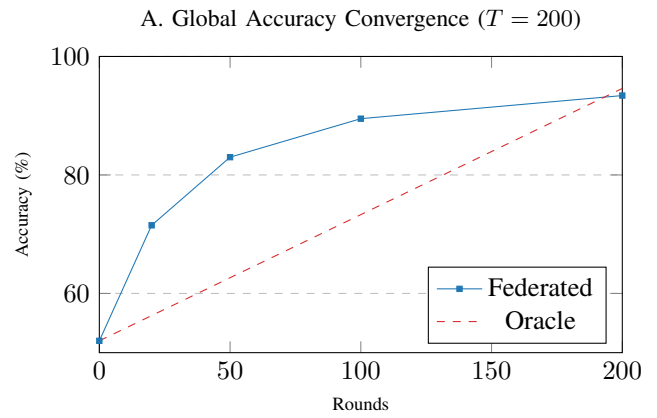


Figure 2. Graph presenting the Global accuracy.

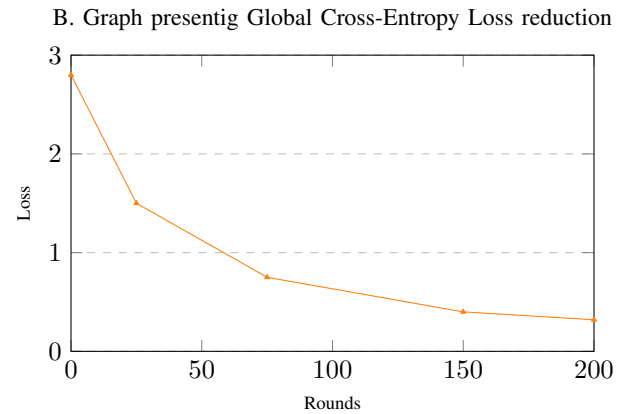


Figure 3. Graph representing Stable cross-entropy loss optimization.

Figure 4 presents that collaborative gain is appreciated among state of art approaches. Furthermore, a standard isolated training module shows that the diagnostic capabilities of nodes unique to a particular pathology (such as Node 2, which deals with retinal detachments and retinitis pigmentosa) were severely affected. The subject nodes that had increased their responsibilities, screening were very poor in detecting these rare conditions; the sensitivity to identify retinitis pigmentosa dropped to just 38% in completely local conditions. In contrast, our proposed federated approach improved node sensitivity by 51% which is in line with the global health equity targets. This enhancement allows the high level specialization service to be successfully transferred to rural clinics through encrypted gradient exchange, which in turn will democratize high-quality access to healthcare.

Figure 4 also highlights the ability of our proposed model to differentiate finegrained multimodal morphologies. Difficult cases like healthy status and myopia were significantly more accurately diagnosed (F scores of 0.98 and 0.96, respectively) and central serous chorioretinopathy (CSCR) and macular scars were still found to have a significant diagnostic value. Although the aggregation process would otherwise annul these fine fluidic spots and tissue changes, the use of CLAHE-

Fig 4. Sensitivity Comparison

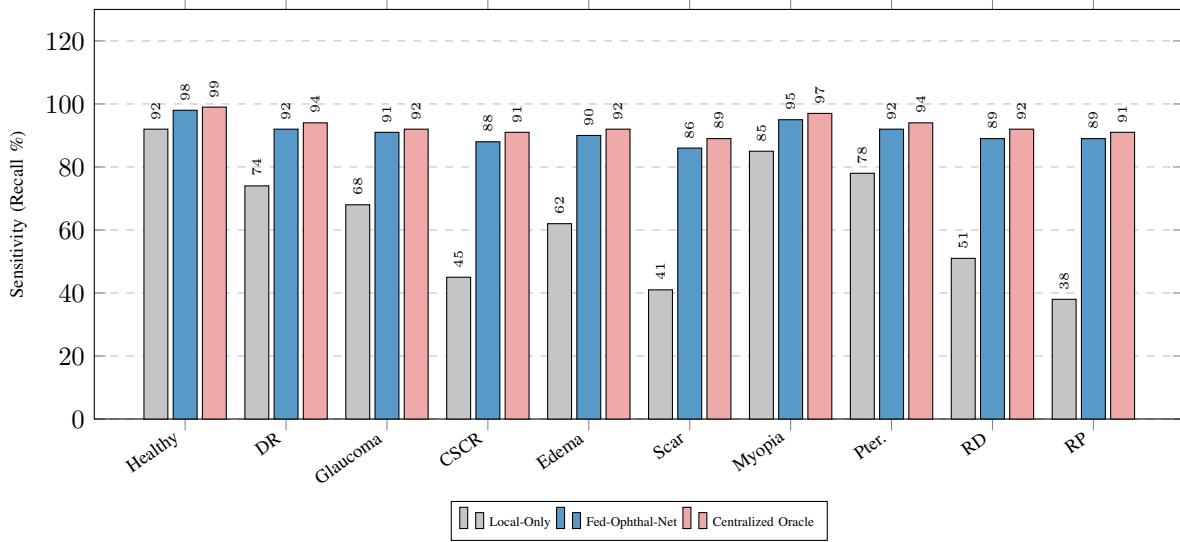


Figure 4. Collaborative Gain and Triple Paradigm Analysis.

Fig 5. ROC Curves: Diagnostic Power

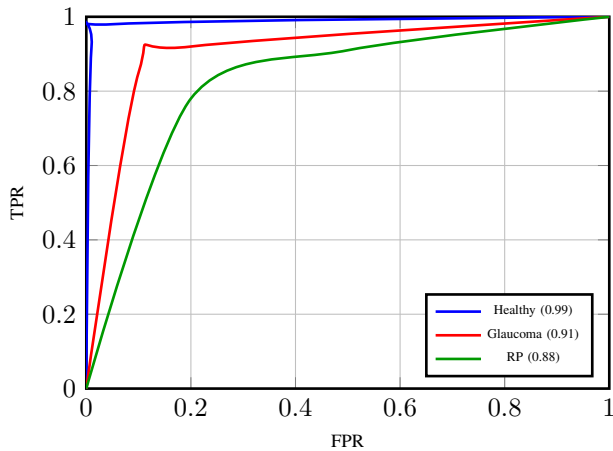


Figure 5. Analysis of Receiver Operating Characteristic (ROC).

enhanced preprocessing and weighted focal loss direction ensured that these subtle aspects were not lost. Figure 5 ROC analysis confirms that even when the balance in the class becomes a critical level.

Figure 3, presents that the loss is steadily decreasing and that training implicitly presupposes some levels of uniformity. The differences between the high-resolution desktop fundus cameras and portable smartphone based devices can cause artifacts, which will be perceived by the existing aggregation logic. Further, weight updates are a limiting condition to clinical uptake; the non-traceability of the explanations of the model behavioural changes is a valid issue. Therefore, the main focus of future researchers should be on incorporating hardware-neutral layers and federated explainable AI (XAI), as well as provisioning heatmaps readable by clinicians and

centralised predictions.

VI. CONCLUSION

The primary objective of this study was to bridge the gap between patient privacy and high-performance eye disease diagnosis. The proposed FedOphthalNet model was implemented and tested in ten different ocular pathologies, demonstrating that decentralization approach is feasible and acceptable compared to the centralised one. Our proposed model achieved an accuracy of 93.4% on a global scale and an increase of 51% in sensitivity to rare diseases. We have determined that the most specialised centres in the diagnosis can be shared with the resource-strapped clinics through the networks of privacy. This research advances the understanding of Federated Learning as a tool for global health equity. It dismantles the barrier of the "data silo" while respecting the sanctity of patient data sovereignty. Ultimately, this study serves as a foundational step toward a truly global, privacy-first diagnostic network that brings expert-level ophthalmic care to every patient, regardless of their geographical location or the local prevalence of their disease.

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